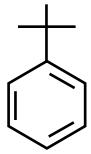
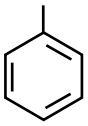
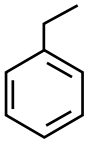
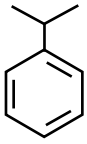
RACE
Regular Analysis Through Continuous Exercise

Aromatic Compounds : RACE-01

1. Which of the following is the correct order of the rate of reaction of C_6H_6, C_6D_6 and C_6T_6 towards nitration?
(A) $C_6H_6 > C_6D_6 > C_6T_6$ (B) $C_6H_6 = C_6D_6 = C_6T_6$
(C) $C_6H_6 > C_6D_6 = C_6T_6$ (D) $C_6T_6 > C_6D_6 > C_6H_6$

2. Which of the following will undergo sulphonation at fastest rate ?
(A)  (B)  (C)  (D) 

3. In a reaction of C_6H_5Y, the major product (>60%) is m-isomer, so the group Y is:
(A) –COOH (B) –Cl (C) –OH (D) $-NH_2$

4. $C_6H_6 \xrightarrow[AlCl_3]{CH_3COCl} A \xrightarrow[HCl]{Zn-Hg} B$
The end product in the above sequence is:
(A) Toluene (B) Ethyl benzene (C) Both the above (D) None

5. For preparing monoalkyl benzene, acylation process is preferred than direct alkylation because
(A) In alkylation, a poisonous gas is evolved
(B) In alkylation, large amount of heat is evolved
(C) In alkylation, polyalkylated product is formed
(D) Alkylation is very costly

Bansal's Academy JEE

6. Chloro atom in chlorobenzene is ortho para director because ?
(A) Resonance effect predominates over inductive effect
(B) Inductive effect predominates over resonance effect
(C) Both inductive effect and resonance effect are evenly matched
(D) Only resonance effect operates

7. Chloro benzene after reaction with molten NaOH undergoes acidification, the compound formed would be:
(A) Phenol (B) Benzoic acid (C) Benzene (D) Disodium benzaldehyde

8. Which mechanism among the following mechanism is followed in Dow's process?
(A) Benzyne mechanism (B) S_NAr
(C) S_N2 (D) S_N1

9. What is the correct IUPAC name of catechol?
(A) Benzene-1,3-diol (B) Benzene-1,4-diol (C) Benzene-1,2-diol (D) None of the above

10. What is the correct IUPAC name of Anisole ?
(A) Methyl phenyl ether (B) Ethyl phenyl ether
(C) Ethoxybenzene (D) Methoxybenzene

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	B	A	B	C	A	A	A	C	D

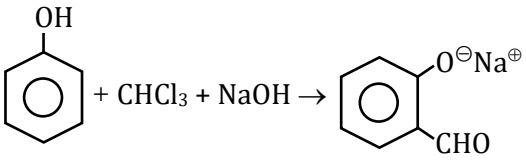
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RACE Regular Analysis Through Continuous Exercise

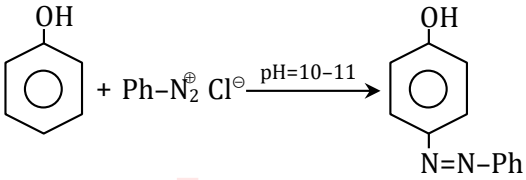
Aromatic Compounds : RACE-02

1. Which of the following compounds shows intramolecular hydrogen bonding :-
(A) p–Nitrophenol (B) Ethanol (C) o–Nitrophenol (D) Methanamine

2. Phenol on treatment with methyl chloride in the presence of anhydrous $AlCl_3$ gives chiefly :-
(A) o–cresol (B) m–cresol (C) anisole (D) p–cresol

3. 

The electrophile involved in the above reaction is
(A) dichlorocarbene ($:CCl_2$) (B) trichloromethyl anion ($\overset{\ominus}{(CCl_3)}$)
(C) formyl cation ($\overset{\oplus}{(CHO)}$) (D) dichloromethyl cation ($\overset{\oplus}{(CHCl_2)}$)

4. 

Characteristics of above reaction is:
(A) C — N coupling reaction ; Carbocation is intermediate
(B) N — N coupling reaction ; Carbocation is intermediate
(C) C — N coupling reaction ; Carbanion is intermediate
(D) N — N coupling reaction ; Carbanion is intermediate

5. Benzene diazo salt give coupling reaction with α – naphthol in alkaline medium, colour of coupling product is :-
(A) Red colour (B) Green colour (C) White colour (D) Blue colour

6. Phenyl acetate on heating in presence of anhydrous $AlCl_3$ gives O-and p-hydroxy acetophenone. This reaction is known as:
(A) Buchner reaction (B) Gattermann is reaction (C) Fries rearrangement (D) Dow process

7. Two aromatic compounds having formula C_7H_8O which are easily identifiable by $FeCl_3$ solution test (violet colouration) are :
(A) Methyl phenyl ether and benzyl alcohol (B) o-cresol and p-cresol
(C) m-cresol and p-cresol (D) o-cresol and benzyl alcohol

8. Phenol and ethanol are distinguished by the reaction with
(A) Red litmus (B) $NaHCO_3$ (C) $FeCl_3$ (D) Na

9. In the Liebermann's nitroso reaction, sequential changes in the colour of phenol occurs as:
(A) Green ⟶ Red ⟶ deep blue (B) Red ⟶ deep blue ⟶ green
(C) Red ⟶ green ⟶white (D) White ⟶ red ⟶ green

ANSWER KEY

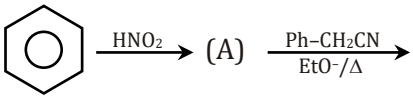
Question	1	2	3	4	5	6	7	8	9
Answer	C	D	A	C	A	C	D	C	A

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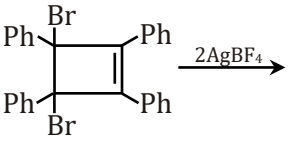
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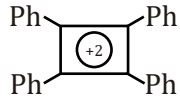
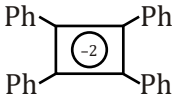
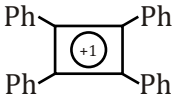
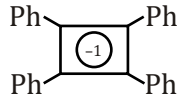
RACE
Regular Analysis Through Continuous Exercise

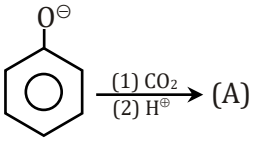
Aromatic Compounds : RACE-03

1. 
; Product (B) is :

(A) Ph–N=C(Ph)–CN (B) Ph–N=C–Ph (C) Ph–N=N–Ph (D) Ph–CH=CH–Ph

2. 
; compound (x) will be :

(A)  (B)  (C)  (D) 

3. 

Which of the following is true statement about the reaction?

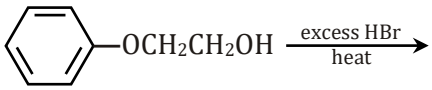
(A) Ortho isomer is major if PhON is used

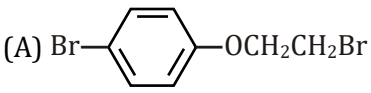
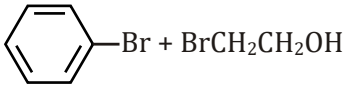
(B) Para isomer is major if PhOK is used

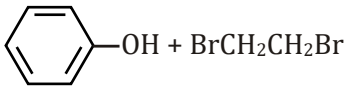
(C) Product formed is further used for preparation of drug aspirin

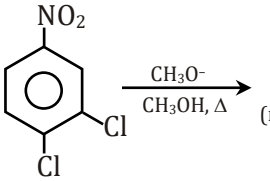
(D) All of these

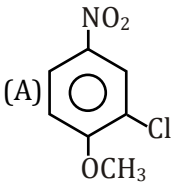
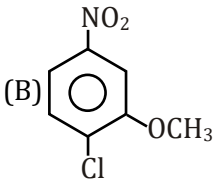
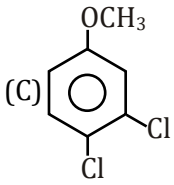
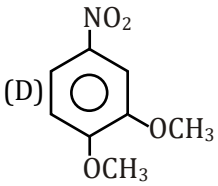
4. What are the products of the following reactions?



(A)  (B) 

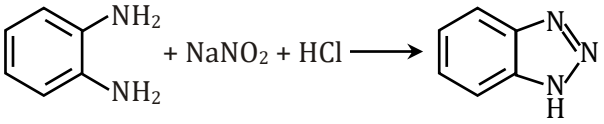
(C)  (D) 

5. 
(P) (major) ; The product (P) will be :

(A)  (B)  (C)  (D) 

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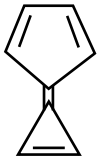
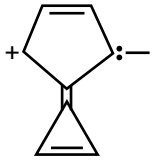
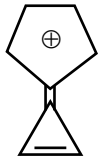
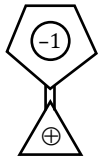
6. 

This reaction is example of :

(A) Intermolecular C–N coupling (B) Intramolecular C–N coupling

(C) Intermolecular N–N coupling (D) Intramolecular N–N coupling

7. Caliene, C_7H_6, is expected to be a fairly polar aromatic molecule. Which of the following resonance forms contributes to the greatest extent towards the real structure (resonance hybrid) of the molecule?

(A)  (B)  (C)  (D) 

8. Consider the following sequence of reactions.

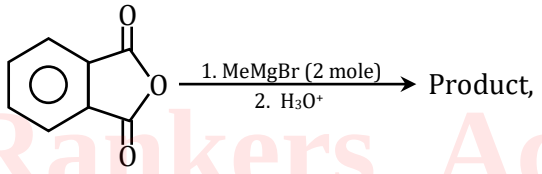
$C_6H_6 + CH_3CH{=}CH_2 \xrightarrow[\text{heat}]{H_3PO_4} A \xrightarrow[2.\ H_3O^+,\ \text{heat}]{1.\ O_2,\ \text{heat}} B + C$

The products (B) and (C) are :

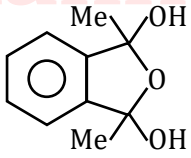
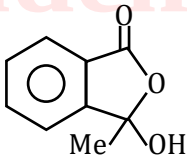
(A) benzaldehyde and acetaldehyde (B) benzoic acid and acetic acid

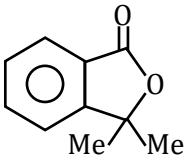
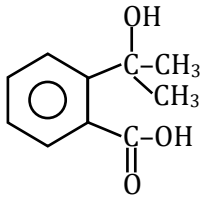
(C) phenol and propionaldehyde (D) phenol and acetone

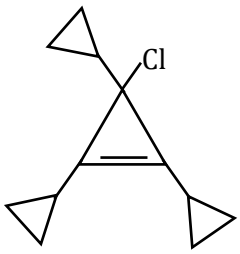
9. For the reaction, the product expected is:



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(A)  (B) 

(C)  (D) 

10. 

In the above compound Cl will liberated easily in the form of :

(A) $Cl^{\oplus}$ (B) $Cl^{\ominus}$ (C) $Cl^{\bullet}$ (D) Cl^{2+}

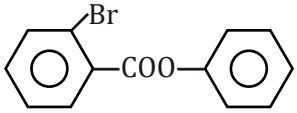
11. Consider the following sequence of reactions :

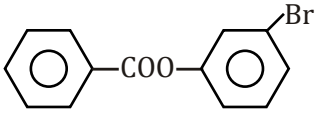
$PhCO_2H \xrightarrow[2.NH_3]{1.PCl_5} A \xrightarrow[2.H_2/Ni]{1.P_4O_{10}.\text{heat}} B$. The final product (B) is :

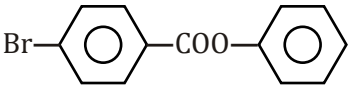
(A) benzonitrile (B) benzylamine (C) aniline (D) benzamide

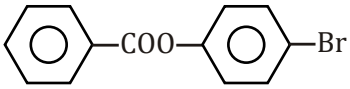
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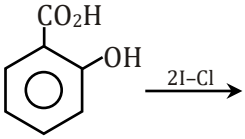
12. The major product expected from the mono–bromination of phenyl benzoate is:

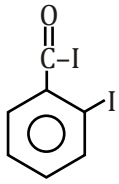
(A) 

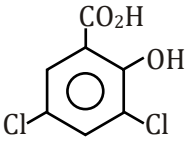
(B) 

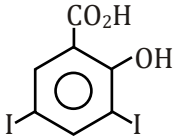
(C) 

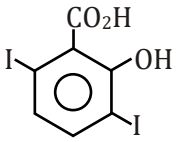
(D) 

13.  ; unknown (A) is :

(A) 

(B) 

(C) 

(D) 

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	A	D	C	A	D	D	D	D	B
Question	11	12	13							
Answer	B	D	C							

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